

Practical No. 03

ON

SSCA4030 - Image Processing

TITLE: Read an 8-bit image and then apply different image enhancement techniques

**BACHLEOR OF SCIENCE IN INFORMATION TECHNOLOGY
(HONOURS.)**

SUBMITTED BY:

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Aim: To understand and implement the conversion of a colour image into grayscale and various colour models such as RGB, HSV, LAB, and YCrCb using Python and OpenCV.

Input Code:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Read 8-bit grayscale image
img = cv2.imread("/content/Flower.jpg", cv2.IMREAD_GRAYSCALE)

# (a) Brightness Improvement
bright = cv2.add(img, 50)

# (b) Brightness Reduction
dark = cv2.subtract(img, 50)

# (c) Thresholding
_, thresh = cv2.threshold(img, 128, 255, cv2.THRESH_BINARY)

# (d) Negative Image
negative = 255 - img

# (e) Log Transformation
img_float = img.astype(np.float32)

c = 255 / np.log(1 + 255)

log_image = c * np.log(1 + img_float)

log_image = cv2.normalize(
    log_image,
    None,
    0,
```

```
255,
cv2.NORM_MINMAX
)

log_image = np.uint8(log_image)

# (f) Power Law (Gamma)
gamma = 0.5

power = np.power(img.astype(np.float32) / 255.0, gamma)
power = np.uint8(power * 255)

titles = [
    "Original",
    "Brightness +",
    "Brightness -",
    "Threshold",
    "Negative",
    "Log",
    "Power Law"
]

images = [
    img,
    bright,
    dark,
    thresh,
    negative,
    log_image,
    power
]

plt.figure(figsize=(14,8))

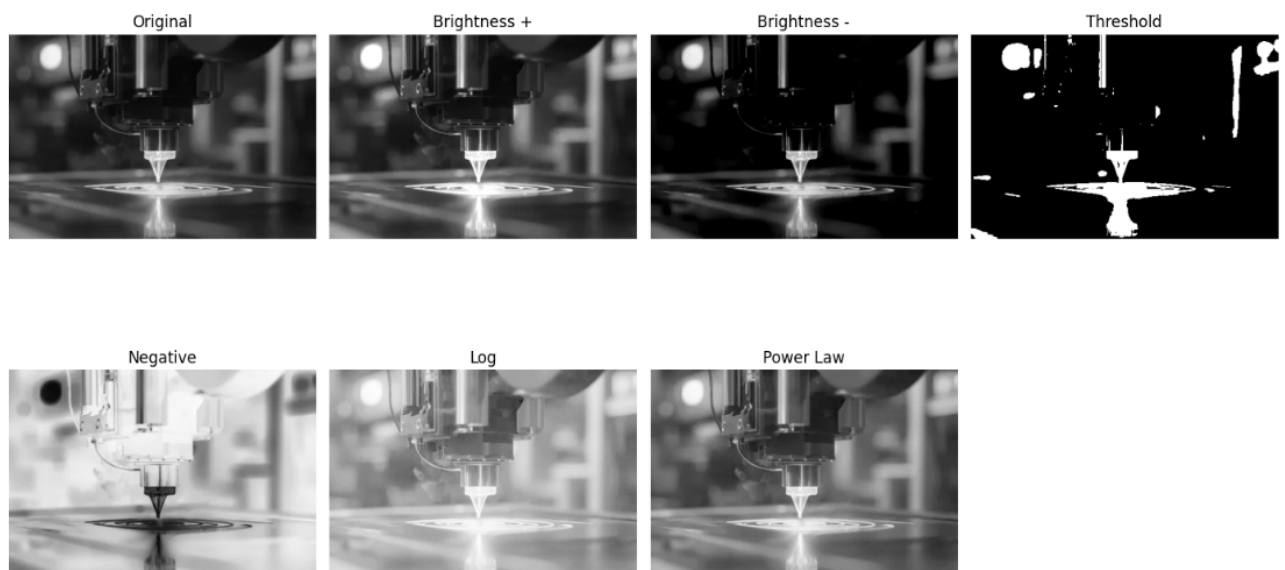
for i in range(len(images)):
```

```
plt.subplot(2,4,i+1)
plt.imshow(images[i], cmap="gray")
plt.title(titles[i])
plt.axis("off")
```

```
plt.tight_layout()
```

```
plt.show()
```

Output:



Formula Used

(a) Brightness Improvement

$$s=r+C$$

where

r = Original pixel

C = Positive constant

(b) Brightness Reduction

$$s=r-C$$

(c) Thresholding

$$g(x,y)=\begin{cases} 255, & f(x,y)\geq T \\ 0, & f(x,y)<T \end{cases}$$

where

T = Threshold value (e.g., 128)

(d) Negative Image

$$s = L - 1 - r$$

For an 8-bit image

$$L = 256$$

Hence,

$$s = 255 - r$$

(e) Log Transformation

$$s = c \log(1 + r)$$

where

$$c = 255 / \log(1 + \text{Maximum Pixel})$$

(f) Power Law (Gamma) Transformation

$$s = cr^\gamma$$

or in normalized form,

$$s = 255(r/255)^\gamma$$

where

$\gamma < 1 \rightarrow$ Bright image

$\gamma > 1 \rightarrow$ Dark image

$\gamma = 1 \rightarrow$ Original image

If Gamma = 0.5 the image becomes brighter.

If Gamma = 2.0 the image becomes darker.